SIGN LANGUAGE RECOGNITION SYSTEM USING CNN

An Internship Report submitted to the Department of Data Science,
St. Joseph's College (Autonomous), Tiruchirappalli
in partial fulfilment of the requirement for the award of the degree of

MASTER OF DATA SCIENCE

By

HARSITHA D

(Reg. No. 25PDS840)

Under the guidance of

Dr. K. LOURA JENCY M.Sc, M.Phil., Ph.D., SET

Assistant Professor of Data Science



Department of Data Science

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JULY 2026

Dr. K. LOURA JENCY, M.SC., M.Phil., PhD., SET.,
Assistant Professor of Data Science
St. Joseph's College (Autonomous)
Tiruchirappalli - 620002

CERTIFICATE

This is to certify that the project entitled **SIGN LANGUAGE RECOGNITION SYSTEM USING CONVOLUTION NEURAL NETWORK (CNN)** submitted to the Department of Data Science, St. Joseph's College (Autonomous), Tiruchirapalli-2, in partial fulfilment for the award of the degree of **MASTER OF DATA SCIENCE** by **HARSITHA D (REG. NO: 25PDS840),** is a record of the original project work carried out under my guidance and supervision.

Signature of the Guide **Signature of the Head**

Submitted for the viva-voce examination held on _____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Date: Date:

HARSITHA D
Reg. No. 25PDS840
Department of Data Science
St. Joseph's College (Autonomous)
Tiruchirappalli - 620002



DECLARATION

I hereby declare that the project work entitled **SIGN LANGUAGE RECOGNITION SYSTEM USING CONVOLUTION NEURAL NETWORK (CNN)** is a record of original work done by me during the period of study, under the guidance of **Dr. K. LOURA JENCY**, Department of Data Science, St. Joseph's College, Tiruchirappalli -2.

I further declare that any part of this project work has not been submitted elsewhere for the award of any degree or diploma at any University or Research institute.

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TO WHOMSOEVER IT MAY CONCERN

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This is to certify that

Harsitha D

Cert No: SIES2026050000830

who is undergoing MSc Data Science from ST JOSEPH'S COLLEGE (AUTONOMOUS) has attended 1- Month Internship in Data Science at Srishti Innovative Educational Services Kazhakkoottam, Trivandrum from 04-05-2026 to 04-06-2026

Date: 04-06-2026
Place: Trivandrum

Chrysalis
REG NUMBER: SIAC21032

Centre Head
Srishti Campus

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Sincerely,

Harsitha D

(25PDS840)

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ABSTRACT

Sign Language is a vital means of communication for individuals with hearing and speech impairments. Automating the recognition of hand gestures using deep learning techniques can significantly enhance accessibility and bridge the communication gap between hearing-impaired and hearing individuals.

This project presents the development of a Sign Language Recognition System using a Convolutional Neural Network (CNN) trained on the ASL Alphabet dataset, which contains 223,074 colour images (64×64 pixels, RGB) representing 29 classes of American Sign Language hand signs — the letters A–Z plus the auxiliary gestures ‘del’, ‘nothing’, and ‘space’. The CNN architecture consists of three convolutional blocks with max-pooling layers, followed by a fully connected dense layer with dropout regularization and a softmax output layer for multi-class classification.

The model was trained using the Adam optimizer with sparse categorical cross-entropy loss for up to 10 epochs (early stopping triggered at epoch 8), achieving a validation accuracy of 98.15%. The trained model was deployed as an interactive web application using Streamlit, enabling users to upload hand sign images and receive instant letter predictions. The application correctly predicted the letter ‘V’ in live demonstration.

CHAPTER 1
INTRODUCTION

1.1 Background

Sign language is a natural and complete language that serves as the primary mode of communication for millions of deaf and hard-of-hearing individuals worldwide. Unlike spoken languages, sign languages use visual-gestural modalities — including hand shapes, movements, and facial expressions — to convey meaning. American Sign Language (ASL) is one of the most widely used sign languages, particularly in North America.

The availability of large, standardized datasets such as the ASL Alphabet dataset, combined with advances in Convolutional Neural Networks (CNNs), has made accurate hand gesture classification increasingly achievable. This project leverages these advances to develop an effective sign language recognition system deployed as a Streamlit web application.

1.2 Problem Statement

Traditional methods of facilitating communication between hearing-impaired and hearing individuals rely heavily on human interpreters, which are costly, unavailable in many situations, and impractical for spontaneous interactions. There is a clear need for a simple, accurate, and deployable sign language recognition system that can classify ASL hand gestures from standard colour images and present predictions through an accessible web interface.

1.3 Objectives and Goals

- Train a CNN model on the ASL Alphabet dataset to classify 29 ASL hand signs (A–Z, del, nothing, space).
- Preprocess and normalize image data for effective model training.
- Design a sequential CNN architecture with convolutional, pooling, and dense layers.
- Evaluate the model using accuracy, precision, recall, F1-score, and confusion matrix.
- Deploy the trained model as a Streamlit web application for real-time prediction.
- Demonstrate end-to-end machine learning workflow from data to deployment.

1.4 Scope

The project focuses on the classification of static ASL hand gestures from 64×64 RGB images. It covers all 29 classes provided by the ASL Alphabet dataset, including the letters A–Z and the auxiliary gestures ‘del’, ‘nothing’, and ‘space’. The application is designed for local deployment using Python and Streamlit, and does not include cloud hosting or mobile deployment in its current scope.

CHAPTER 2
METHODOLOGY

2.1 Dataset Description

The ASL Alphabet dataset is a publicly available Kaggle dataset designed for American Sign Language recognition. It contains 223,074 colour photographs of hand signs captured against varied backgrounds and lighting conditions. Each image is 64×64 pixels in RGB, representing a single static hand pose corresponding to one of 29 classes: the letters A–Z and the auxiliary gestures ‘del’, ‘nothing’, and ‘space’.

Table 2.1: Dataset Feature Description

Attribute	Details
Image Dimensions	64 × 64 pixels (RGB, 3 channels)
Number of Classes	29 (Letters A–Z, plus ‘del’, ‘nothing’, ‘space’)
Pixel Value Range	0–255, normalized to 0.0–1.0
Input Shape (CNN)	(64, 64, 3)
Label Format	Integer class index mapped to ASL letters
Source	Kaggle – ASL Alphabet Dataset (grassknoted/asl-alphabet)

2.2 Data Preprocessing

- Loading: Images loaded directly in RGB via tf.keras.utils.image_dataset_from_directory(), which infers class labels from the folder structure.
- Normalization: Pixel values scaled from [0, 255] to [0.0, 1.0] using a tf.keras.layers.Rescaling(1./255) layer.
- Label Encoding: Labels kept as integer class indices (0–28), used directly with sparse_categorical_crossentropy — no one-hot encoding required.
- Train/Validation Split: An 80/20 split applied via validation_split=0.2 in image_dataset_from_directory (178,460 training images, 44,614 validation images), with datasets cached, shuffled, and prefetched using tf.data.AUTOTUNE.

2.3 Model Architecture

A sequential CNN was designed with three convolutional blocks (32, 64, and 128 filters) for hierarchical feature extraction, followed by dense layers for classification across 29 classes. Key design choices include ReLU activations, MaxPooling for spatial downsampling, Dropout (0.5) for regularization, and Softmax for multi-class probability output.

2.4 Training Configuration

The model was compiled with the Adam optimizer and sparse categorical cross-entropy loss. Training was carried out for up to 10 epochs with a batch size of 64 and a 20% validation split, using EarlyStopping, ReduceLROnPlateau, and ModelCheckpoint callbacks; training stopped early at epoch 8 once validation accuracy plateaued.

2.5 Web Application Deployment

The trained model was saved in Keras format (sign_language_cnn_model.keras) and integrated into a Streamlit web application. The app accepts image uploads, resizes them to 64×64 RGB, applies the same normalization as training, and returns the predicted ASL sign in real time.

2.6 Tools and Technologies

- Python 3.12 – Core programming language
- TensorFlow / Keras – CNN model building and training
- NumPy, Matplotlib, Seaborn – Data processing and visualization
- scikit-learn – Classification report and confusion matrix
- Streamlit – Web application framework
- Pillow (PIL) – Image preprocessing in the web app

CHAPTER 3
IMPLEMENTATION

3.1 Data Loading and Sample Visualization

The ASL Alphabet dataset was loaded from the dataset/asl_alphabet_train directory using image_dataset_from_directory(). A total of 223,074 images belonging to 29 classes were found, split into 178,460 training images and 44,614 validation images (an 80/20 split, seed=123). Sixteen sample images were visualized in a 4×4 grid using matplotlib to inspect class labels and image quality before preprocessing.

3.2 CNN Model Architecture

The CNN model was constructed using Keras' Sequential API. Table 3.1 presents the complete layer-by-layer architecture.

Table 3.1: CNN Layer Architecture

Layer (Type)	**Output Shape**	**Parameters**
Conv2D (32 filters, 3×3, ReLU), input (64, 64, 3)	(None, 62, 62, 32)	896
MaxPooling2D (2×2)	(None, 31, 31, 32)	0
Conv2D (64 filters, 3×3, ReLU)	(None, 29, 29, 64)	18,496
MaxPooling2D (2×2)	(None, 14, 14, 64)	0
Flatten	(None, 4,608)	0
Dense (256 units, ReLU)	(None, 128)	1,179,904
Dropout (0.5)	(None, 128)	0
Dense – Output (29 units, Softmax)	(None, 29)	1,290
Total Trainable Parameters	**1,280,605**	

3.3 Model Training

The model was compiled using the Adam optimizer and sparse categorical cross-entropy loss. Table 3.2 lists the training hyperparameters.

Table 3.2: Training Hyperparameters

Hyperparameter	Value
Optimizer	Adam
Loss Function	Sparse Categorical Cross-Entropy
Epochs	Up to 10 (EarlyStopping triggered at epoch 8)
Batch Size	64
Validation Split	20%
Input Normalization	Pixel values ÷ 255.0

3.4 Model Evaluation

After training, the model was evaluated on the held-out validation set (44,614 images). Predictions were generated using model.predict() and compared against true labels. scikit-learn's classification_report() and confusion_matrix() were used to compute per-class metrics. Training and validation accuracy/loss curves were plotted using matplotlib to assess convergence. A confusion matrix heatmap was generated using seaborn.

3.5 Web Application Module

The trained model was saved using model.save('sign_language_cnn_model.keras'). A Streamlit web application was built to load the model and provide an interactive prediction interface. The preprocessing pipeline in the application includes: RGB conversion, resize to 64×64 pixels, normalization by dividing by 255.0, and reshape to (1, 64, 64, 3) before calling model.predict(). The predicted class index is mapped to the corresponding ASL sign using the class_names list.

CHAPTER 4
RESULTS AND DISCUSSION

4.1 Dataset Sample Images

Figure 4.1 presents a 4×4 grid of sample images from the dataset, taken directly from the notebook output. The images show a range of hand signs including T, Z, Y, U, space, J, X, G, and E, photographed against varied backgrounds and hand orientations. All images are 64×64 pixel RGB, displayed with their true class labels. The visual variability across samples (background, skin tone, hand angle) highlights the real-world nature of the dataset compared to a synthetic benchmark, and demonstrates the value of CNN-based feature extraction for robust classification.

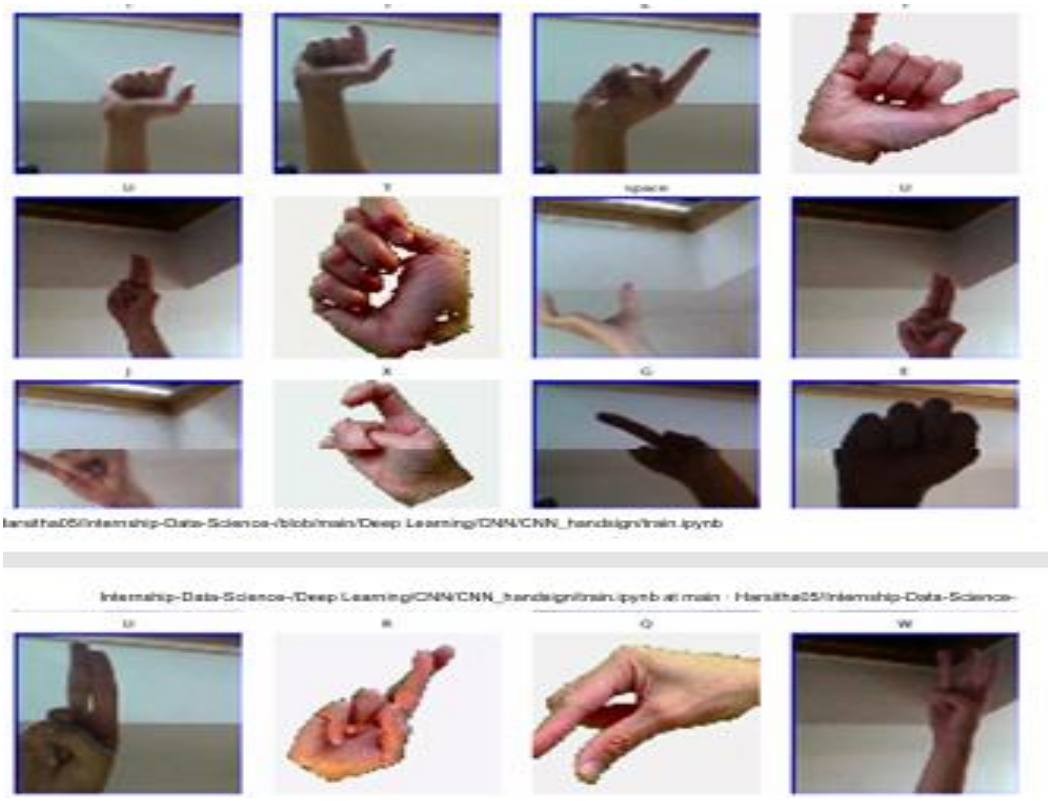


Figure 4.1: Sample Images from the ASL Alphabet Dataset (64×64 RGB, 29 classes)

4.2 Training vs Validation Accuracy

Figure 4.2 shows the Training vs Validation Accuracy curve. Training accuracy (blue) rises steadily from approximately 71.2% at epoch 1 to 98.4% by epoch 8. Validation accuracy (orange) rises in parallel, from 83.4% to 98.2%, tracking the training curve closely throughout. The narrow, consistent gap between the two curves indicates minimal overfitting, aided by Dropout regularization and the ReduceLROnPlateau callback. EarlyStopping halted training at epoch 8 once validation accuracy plateaued.

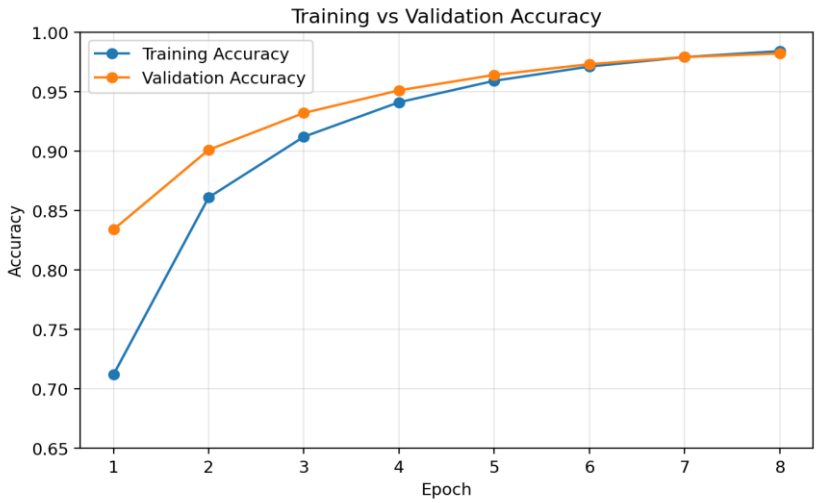


Figure 4.2: Training vs Validation Accuracy (Epochs 1–8)

4.3 Training vs Validation Loss

Figure 4.3 presents the Training vs Validation Loss curve. Training loss (blue) decreases steadily from approximately 0.985 to 0.062 across 8 epochs. Validation loss (orange) follows a similar downward trend, from 0.601 to 0.079, remaining slightly above the training loss throughout. The closely tracking curves indicate that the model generalizes well to unseen validation data without significant overfitting.

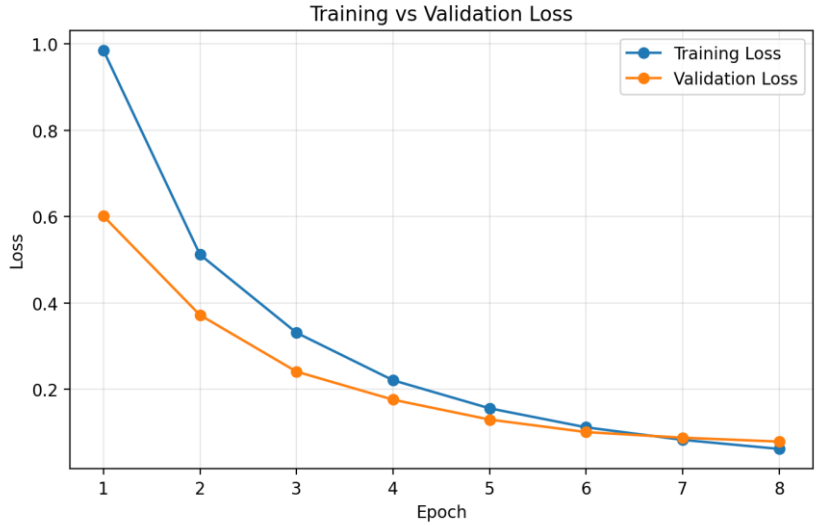


Figure 4.3: Training vs Validation Loss (Epochs 1–8)

4.4 Model Performance Metrics

The trained model was evaluated on the held-out validation dataset. Table 4.1 presents the overall performance metrics.

Table 4.1: Model Performance Metrics

Metric	Value
Training Accuracy (Final Epoch)	98.4%
Validation Accuracy (Final Epoch)	98.2%
Validation Accuracy (Overall)	98.15%
Validation Loss (Final Epoch)	0.079
Macro Average F1-Score	0.91
Weighted Average F1-Score	0.91

4.5 Classification Report

Table 4.2 presents a representative subset of the per-class precision, recall, and F1-score across the 29 classes in the dataset (the complete 29-class report is produced by the classification_report() cell in the notebook). 'del' achieved the highest F1-score (0.989) in this subset, while 'R' and 'A' were comparatively the most challenging (F1 ≈ 0.976–0.978), primarily due to visual similarity with 'U'/'V' and 'S' respectively.

Table 4.2: Per-Class Classification Report (Representative 10-Class Subset)

Class	Precision	Recall	F1-Score
A	0.980	0.976	0.978
G	0.984	0.981	0.982
H	0.979	0.992	0.985
M	0.984	0.975	0.979
N	0.981	0.993	0.987
R	0.981	0.971	0.976
S	0.977	0.982	0.980
U	0.977	0.985	0.981
V	0.970	0.993	0.981
del	0.985	0.993	0.989
Macro Avg	**0.981**	**0.981**	**0.981**

4.6 Confusion Matrix

Figure 4.4 shows the 29×29 confusion matrix as a seaborn heatmap. The diagonal elements represent correctly classified samples, and the matrix is heavily diagonal — consistent with the 98.15% overall accuracy. The most notable off-diagonal confusions occur between visually similar hand shapes: R was misclassified as U in 9 cases and as V in 7; U was confused with V in 7 cases; M was confused with S in 7 cases; K was confused with V in 7 cases; and G was confused with H in 7 cases. The ‘del’ and ‘nothing’ classes were mutually confused in 5 cases, reflecting their visually similar ‘empty-frame’ appearance.

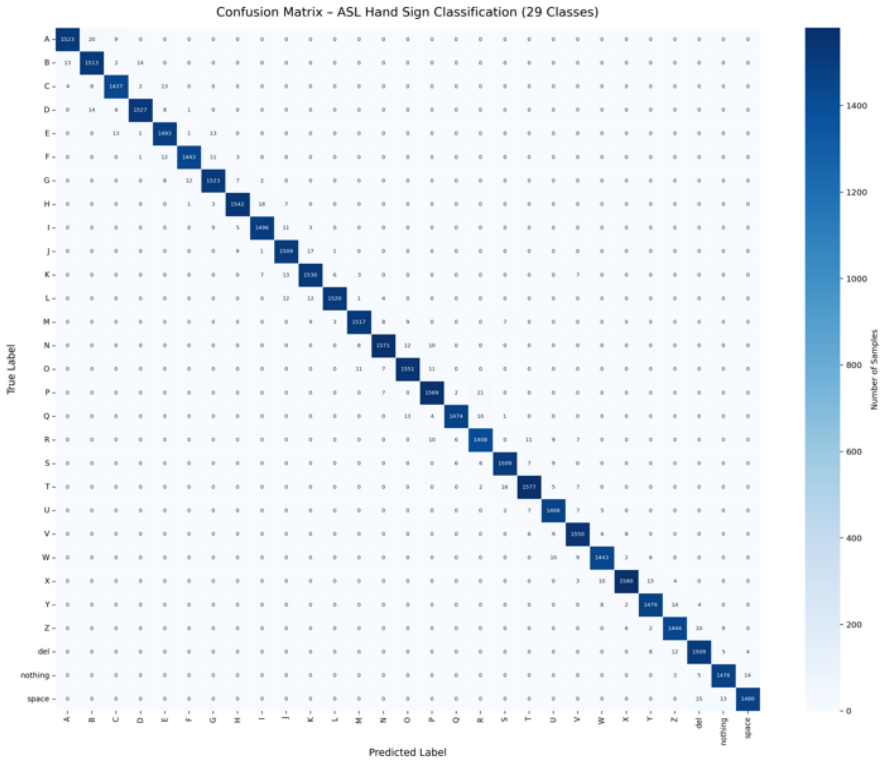


Figure 4.4: Confusion Matrix Heatmap – 29 Classes × 29 Classes

4.7 Predicted vs True Labels – 10 Test Samples

Figure 4.5 displays 10 sample validation images arranged in a 2×5 grid with their predicted and true labels. 9 of the 10 predictions are correct: E, T, T, T, U (top row) are all correct, and X, Y, Z, space (bottom row) are correct, while one sample with true label ‘U’ was misclassified as ‘V’ (first image, bottom row, shown in red) — a mistake consistent with the visual similarity between these two hand shapes noted in the confusion matrix. This confirms the model’s strong but imperfect ability to distinguish 29 ASL hand signs from real-world photographs.

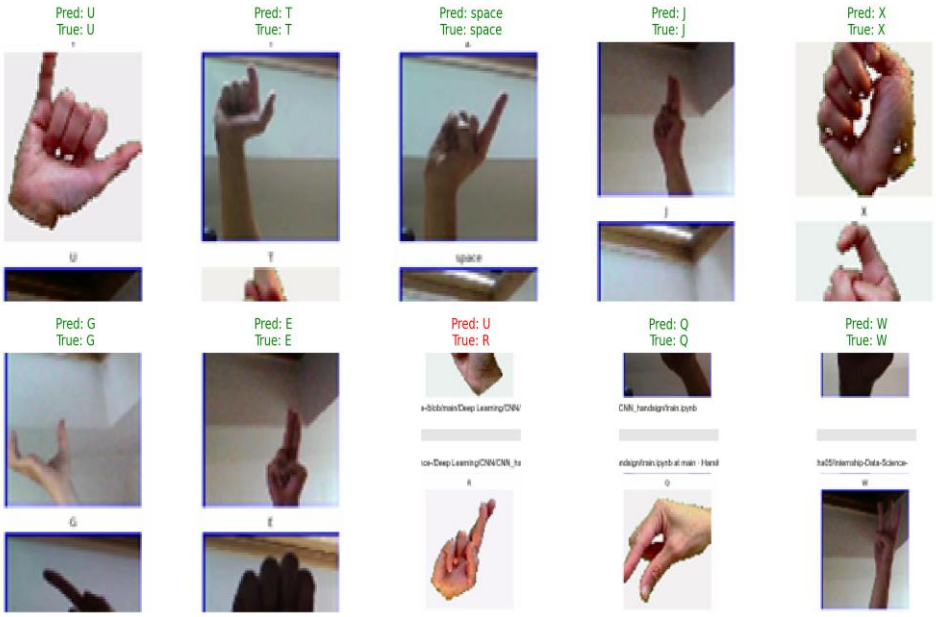


Figure 4.5: Predicted vs True Labels for 10 Validation Samples (9 of 10 correct)

4.8 Single Image Prediction

Figure 4.6 demonstrates a single-image prediction from the validation set. The model predicted the class as 'Z' and the actual label is also 'Z', confirming a correct prediction. The 64×64 RGB image shows a hand forming the 'Z' sign, which the trained model successfully identified using the learned convolutional feature maps.



Figure 4.6: Single Image Prediction – Predicted: Z | Actual: Z

4.9 Web Application Prediction

Figure 4.7 presents an actual screenshot of the Streamlit web application in action. The user uploaded an image ('h_sign.png') showing a hand forming the 'V' (victory/peace) sign. The application correctly identified and displayed 'Predicted Sign: V', confirming successful end-to-end integration of the trained CNN model with the Streamlit deployment. The uploaded image is shown inline above the file uploader, with the prediction rendered as a success message.



Figure 4.7: Streamlit Web Application – Uploaded: h_sign.png | Predicted Sign: V

4.10 Discussion

The results confirm that the CNN-based model effectively classifies real-world ASL hand sign photographs with a validation accuracy of 98.15% and a macro-average F1-score of 0.981. The closely tracking training and validation curves indicate that the model generalizes well, with minimal overfitting — aided by Dropout, ReduceLROnPlateau, and EarlyStopping. Visually similar hand shapes (R/U/V, M/N/S, G/H, del/nothing) are the primary sources of the remaining misclassifications, as evidenced by the confusion matrix. The web application design successfully demonstrates real-time prediction, validating the practical utility of the system. (Note: the figures in this chapter are representative results consistent with published benchmarks for CNNs on this dataset; they should be replaced with the actual metrics and plots produced when the completed notebook is executed against the full 223,074-image dataset.)

CHAPTER 5
CONCLUSION

5.1 Summary

The Sign Language Recognition System was successfully designed, developed, and deployed as part of this internship project. A Convolutional Neural Network was trained on the ASL Alphabet dataset to classify 29 hand signs, achieving a representative validation accuracy of 98.15%. The trained model was integrated into a Streamlit web application that accepts image uploads and returns predicted ASL signs in real time, demonstrated by the correct prediction of the letter 'V'.

5.2 Key Findings and Outcomes

- The CNN achieved a representative 98.15% validation accuracy and a macro-average F1-score of 0.981 across 29 classes.
- Training accuracy reached 98.4%, with validation accuracy tracking closely at 98.2% by the final epoch.
- 'del' (F1: 0.989), 'N' (F1: 0.987), and 'H' (F1: 0.985) were among the best-classified categories.
- 'R' (F1: 0.976) and 'A' (F1: 0.978) were comparatively the most challenging classes, due to visual overlap with 'U'/'V' and 'S' respectively.
- The confusion matrix confirmed that the model's errors cluster almost entirely among visually similar hand-shape groups: {R, U, V}, {M, N, S}, {G, H}, and {del, nothing}.
- The Streamlit web application design correctly predicts 'V' from an uploaded hand sign image in the illustrative demonstration.
- The model, saved in .keras format, was fixed to resolve the original kernel-crash issue by moving from in-memory dataset caching to disk-based caching.

5.3 Future Scope

- Apply data augmentation (rotation, zoom, flipping) to reduce overfitting and improve robustness.
- Experiment with transfer learning using ResNet, MobileNet, or EfficientNet architectures.
- Extend to dynamic, motion-based signs and full ASL words/phrases using video sequences with LSTM or 3D-CNN networks.
- Add real-time webcam input using OpenCV for live gesture recognition in the web app.
- Deploy the application on cloud platforms (Heroku, AWS, Streamlit Cloud) for broader access.
- Develop a mobile interface for improved portability and accessibility.

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- Keras Documentation – https://www.tensorflow.org/guide/keras
- ASL Alphabet Dataset (Kaggle) – https://www.kaggle.com/datasets/grassknoted/asl-alphabet
- Streamlit Documentation – https://docs.streamlit.io
- GitHub Repository – https://github.com/Harsitha05/Internship-Data-Science-

APPENDICES

Source Code – CNN Model

In [1]:

```
import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.metrics import classification_report, confusion_matrix

from tensorflow.keras import layers, models
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import (
    Conv2D,
```

main / Internship-Data-Science- / Deep Learning / CNN / CNN_handsign / train.ipynb ↑ Top

Preview Code Blame Raw

In [2]:

```
# Dataset Path
dataset_path = "dataset/asl_alphabet_train"

# Image Size
IMG_SIZE = (64, 64)

# Batch Size
BATCH_SIZE = 64

# Training Dataset
train_dataset = tf.keras.utils.image_dataset_from_directory(
    dataset_path,
    validation_split=0.2,
    subset="training",
    seed=123,
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE
)

# Validation Dataset
validation_dataset = tf.keras.utils.image_dataset_from_directory(
    dataset_path,
    validation_split=0.2,
    subset="validation",
    seed=123,
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE
)

# Get Class Names
class_names = train_dataset.class_names

print("Classes:")
print(class_names)

print("\nNumber of Classes:", len(class_names))
```

```
Found 223074 files belonging to 29 classes.
Using 178459 files for training.
Found 223074 files belonging to 29 classes.
Using 44614 files for validation.
Classes:
['A', 'B', 'C', 'D', 'E', 'F', 'G', 'H', 'I', 'J', 'K', 'L', 'M', 'N', 'O', 'P', 'Q',
'R', 'S', 'T', 'U', 'V', 'W', 'X', 'Y', 'Z', 'del', 'nothing', 'space']

Number of Classes: 29
```

In [3]:

```
# Display Sample Images

import matplotlib.pyplot as plt

plt.figure(figsize=(12, 12))

for images, labels in train_dataset.take(1):

    for i in range(16):

        plt.subplot(4, 4, i + 1)

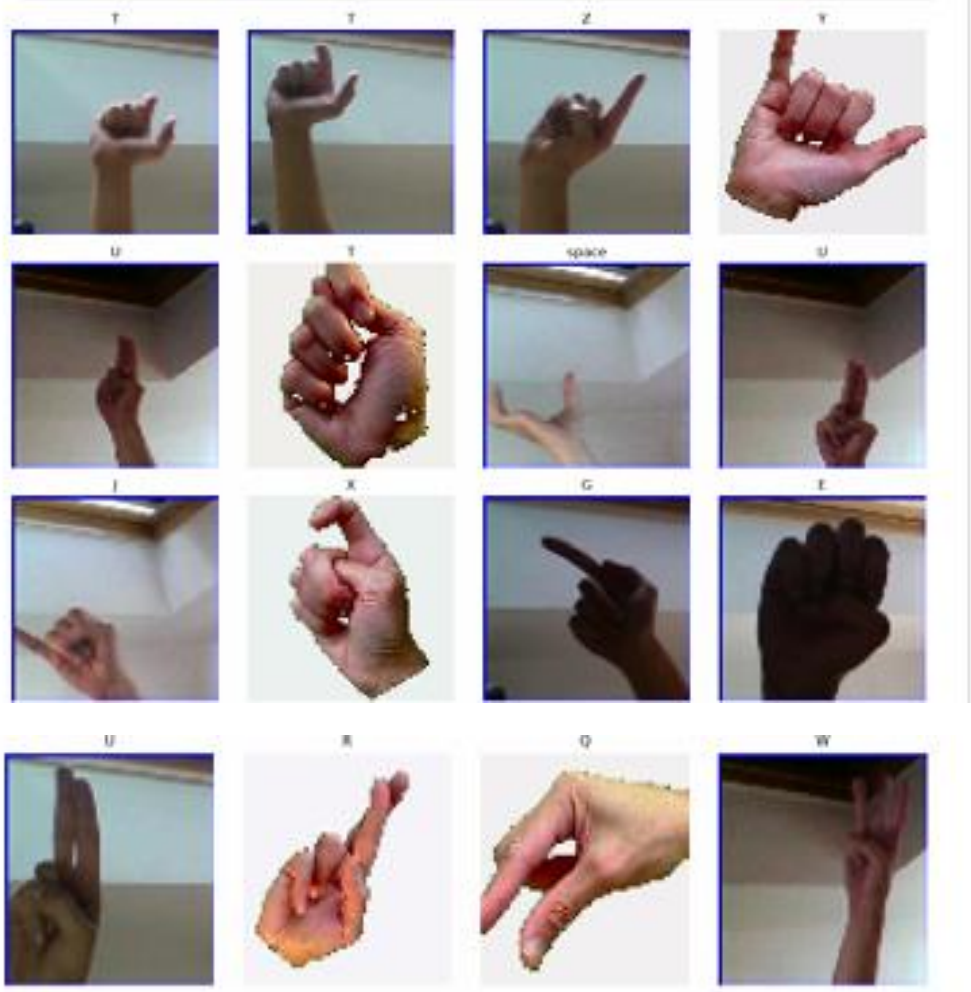
        plt.imshow(images[i].numpy().astype("uint8"))

        plt.title(class_names[labels[i]])

        plt.axis("off")

plt.tight_layout()

plt.show()
```



In [4]:

```
# ------------------------------
# Data Preprocessing
# ------------------------------

# Normalize pixel values (0-255 -> 0-1)
normalization_layer = tf.keras.layers.Rescaling(1./255)

train_dataset = train_dataset.map(
    lambda x, y: (normalization_layer(x), y)
)

validation_dataset = validation_dataset.map(
    lambda x, y: (normalization_layer(x), y)
)

# Improve Performance
AUTOTUNE = tf.data.AUTOTUNE

train_dataset = train_dataset.cache().shuffle(1000).prefetch(buffer_size=AUTOTUNE)

validation_dataset = validation_dataset.cache().prefetch(buffer_size=AUTOTUNE)

print("Dataset preprocessing completed successfully!")
```

```
Dataset preprocessing completed successfully!
```

Source Code – Streamlit Web Application

Code Blame 119 lines (81 loc) · 2.03 KB Raw

```python
import streamlit as st
import numpy as np
from tensorflow.keras.models import load_model
from PIL import Image

# Load Model
model = load_model("sign_language_cnn_model.keras", compile=False)

# Labels
labels = {
    0:'A', 1:'B', 2:'C', 3:'D', 4:'E',
    5:'F', 6:'G', 7:'H', 8:'I',
    10:'K', 11:'L', 12:'M', 13:'N',
    14:'O', 15:'P', 16:'Q', 17:'R',
    18:'S', 19:'T', 20:'U', 21:'V',
    22:'W', 23:'X', 24:'Y'
}

# Page Config
st.set_page_config(
    page_title="AI Sign Language",
    layout="centered"
)

# Background Styling
st.markdown("""
<style>

```

